SURGSTREAM: A WORKFLOW-ANCHORED PROACTIVE STREAMING FRAMEWORK AND BENCHMARK FOR REAL-TIME SURGICAL VIDEO UNDERSTANDING

Ruipeng Ma[1,2], *Qing Xu*[3], *Jie Liang*[2,✉], *Zhen Chen*[1,✉]

[1]Department of Data Science and Artificial Intelligence, The Hong Kong Polytechnic University
[2]Eastern Institute of Technology, Ningbo [3]University of Nottingham, UK

ABSTRACT

Surgical video understanding is essential for intraoperative assistance such as workflow monitoring, instrument preparation and procedural documentation. Delivering such assistance in real time requires a model to jointly perceive the ongoing scene, remember past surgical events, and respond before clinically relevant moments pass. Existing surgical video-language models either operate offline on complete videos or narrate only the current frames, failing to retain procedural evidence and to decide when a response is required. In this work, we propose SurgStream, a workflow-anchored proactive streaming framework for real-time surgical video understanding. We first introduce Workflow-Anchored Memory (WAM) to record structured instrument–action–target observations and fold them into phase-level summaries, providing compact and queryable surgical memory. Deadline-aware Transition Learning (DTL) is then devised to supervise response states within clinically bounded windows, enabling timely responses before procedural deadlines. We further introduce SurgStream-Bench, comprising 11.7K streaming interactions across various tasks covering real-time perception, backward memory queries, forward proactive QA, proactive alerts and caption memory. Experiments on SurgStream-Bench show gains over state-of-the-art methods, especially in memory and forward proactive QA. Project homepage: `https://surgstream.github.io/`.

***Index Terms*—** Surgical video understanding, streaming video LLM, workflow memory, proactive response

1. INTRODUCTION

Real-time surgical video understanding plays a crucial role in intraoperative assistance [1, 2], such as phase-aware instrument preparation, procedural documentation and progress tracking. Such assistance involves understanding the current scene, recalling past events and delivering responses within relevant procedural windows. Figure 1 illustrates these interaction requirements and the temporal modes considered in this work. Surgical videos are long, visually homogeneous and densely structured by the workflow [3]. For time-sensitive assistance, delayed responses may lose their practical value, making real-time surgical interaction challenging.

Early surgical video-language models primarily focused on visual question answering on images or complete videos [4, 5, 6, 7]. Subsequently, general and medical vision-language models [8, 9, 10] achieved significant progress in visual understanding, and real-time surgical commentary [11] narrated the current frames on the fly. More recently, streaming video large language models (LLMs) [12, 13, 14, 15, 16] have enabled causal perception and proactive response on continuous video streams, and OneStreamer [17] further unifies perception, memory and proactive response in a single model.

Despite this progress, existing methods are not explicitly designed around surgical workflows. Free-text memory does not explicitly preserve phase boundaries or structured event counts, making retrospective queries difficult to support reliably. Response supervision at a single annotated time [17] also leaves the interval of acceptable responses implicit. These limitations motivate workflow-structured memory, supervision within valid response windows, and evaluation of both answer content and timing.

To address these limitations, we propose SurgStream, a workflow-anchored proactive streaming framework that integrates compact surgical memory with response supervision within valid procedural windows. For memory construction, we introduce Workflow-Anchored Memory (WAM) to record stable instrument–action–target changes and consolidate each completed phase into a compact *Summary*, preserving evidence for retrospective queries. We further devise Deadline-aware Transition Learning (DTL) to resample response targets within label-derived clinical windows in each epoch, encouraging early responses once evidence is available and before procedural deadlines. Moreover, we construct SurgStream-Bench, spanning diverse tasks across five clinical scenarios and five temporal modes. Extensive experiments demonstrate that SurgStream outperforms competing methods, with particularly strong gains in memory and forward proactive QA.

2. METHODOLOGY

As shown in Fig. 2, SurgStream performs real-time surgical video interaction within a single streaming video LLM. Given a video stream $\{x_t\}_{t=1}^{T}$ arriving at 1 fps, the model reads at each step t the causal context $C_t = [M_t; V_t; D_t]$, consisting of the memory M_t, the recent visual window $V_t = \{x_{t-W+1}, \ldots, x_t\}$ and the dialogue D_t, and emits a control state $s_t \in \{$*Silence*, *Standby*, *Response*, *Observe*, *Summary*$\}$ followed by optional text. On this basis, WAM and DTL anchor what to remember and when to respond to the surgical workflow without new network modules, and SurgStream-Bench evaluates real-time surgical video interaction.

2.1. Workflow-Anchored Memory

To provide compact and queryable surgical memory, we propose WAM, which anchors both the write points and the content of memory to the surgical workflow instead of letting the model decide the boundaries of free-text captions. Let u_t denote the structured surgical state at time t, *i.e.*, the set of instrument–action–target triplets or present instruments. When u_t changes and the new state remains stable for κ seconds, the model writes an *Observe* record $o_\tau = \langle t, \Delta u_t \rangle$ at $\tau = t+\kappa-1$, where Δu_t denotes the elements newly appearing at time t relative to the preceding state. The record lists these elements as "[mm:ss] instrument | verb | target". Let O_k collect the *Observe* records o_τ whose event times t fall within phase k, $[b_k, b_{k+1})$. After the phase boundary b_{k+1} is confirmed, the model writes a *Summary*

✉ Corresponding author.

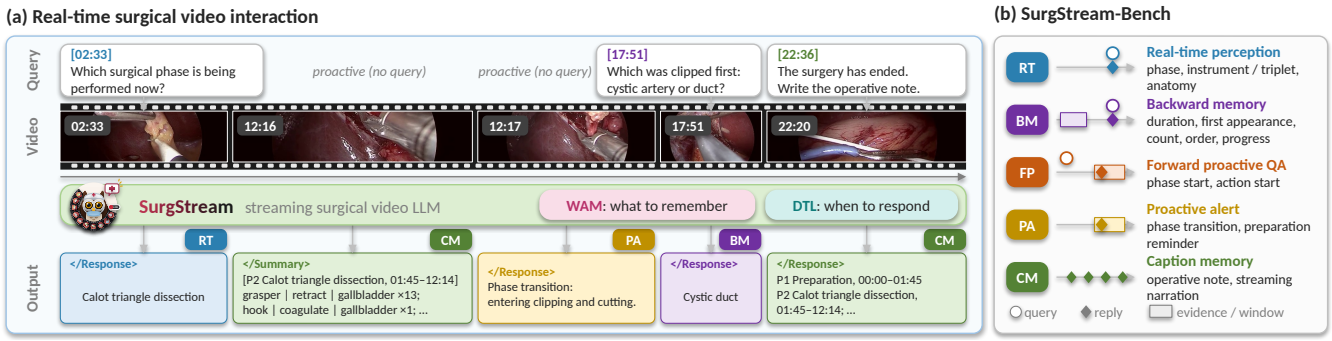


Fig. 1. Illustration of SurgStream and SurgStream-Bench. (a) An example of streaming surgical interaction, including current-scene questions, retrospective queries, proactive alerts and memory writing. (b) The five temporal modes evaluated by SurgStream-Bench.

record $\sigma_k = \langle p_k, [b_k, b_{k+1}), \{(c, n_k(c))\}\rangle$, where p_k is the phase label and $n_k(c)$ counts the onsets of element c in O_k (*e.g.*, clip×3). We replace O_k with σ_k and retain the other memory records R_t:

$$\begin{aligned} R_t &= \{\, m \in M_t \mid m \notin O_k \,\}, \\ M_{t+1} &= R_t \cup \{\sigma_k\}. \end{aligned} \tag{1}$$

In this way, memory grows with the number of phases rather than with surgery duration, and questions on durations, first appearances and counts can be answered directly from the records. Since all records are derived from expert labels, memory writing is supervised by L_{mem}, the token-averaged cross-entropy of each record given its state and causal context. WAM thus provides queryable surgical memory that is verifiable against expert labels and remains compact for long procedures.

2.2. Deadline-aware Transition Learning

To enable timely responses within valid clinical windows, we propose DTL, which extends generic sparse supervision [17] beyond a single annotated response time. Each response target j is assigned a window $[a_j, d_j]$, where the evidence time a_j and deadline d_j are derived from labels (*e.g.*, a preparation reminder is due at the next phase onset). To favour early responses, we resample the target time within the valid window at each epoch using inverse transform sampling. Let $U_j \sim \text{Uniform}(0,1)$ and $\Delta_j = d_j - a_j + 1$. The sampled delay is:

$$\tau_j = -\eta \log\left[1 - U_j \left(1 - e^{-\Delta_j/\eta}\right)\right]. \tag{2}$$

The response time is then:

$$r_j = \min\{d_j,\ a_j + \lfloor \tau_j \rfloor\}. \tag{3}$$

Here, timestamps are discretized at one-second intervals, and $\eta > 0$ controls the decay of the sampling distribution. For $a_j = d_j$, we set $r_j = a_j$. We assign $s^*_{r_j}$ =*Response*, s^*_t =*Standby* for $t \in [a_j - \epsilon, r_j)$, and *Silence* elsewhere. Following [17], the supervised set Ω comprises all output anchors (including the resampled r_j), state transitions and representative state-persistence positions, and the state loss is:

$$L_{\text{DTL}} = -\frac{1}{|\Omega|} \sum_{t \in \Omega} \log \pi_\theta(s^*_t \mid C_t). \tag{4}$$

Keeping Ω sparse prevents the abundant *Standby* positions from dominating the loss and suppressing *Response*. As r_j is redrawn in each epoch with earlier times favoured by Eq. (2), DTL teaches the model not only whether but also when to respond, encouraging earlier responses within valid windows.

SurgStream is trained jointly with the cross-entropy of the reply text, L_{mem} and L_{DTL}. At inference, the states are emitted in the order *Summary*→*Observe*→*Response* until *Silence* or *Standby*, and the context is bounded by the visual window and memory records, independent of surgery duration.

2.3. SurgStream-Bench

We construct SurgStream-Bench by converting expert annotations from Cholec80 [3] and CholecT50 [23] into streaming interactions. Phase, instrument and action-triplet labels are aligned along each procedure. Shared recordings are identified using the official overlap mapping [24], represented once with merged labels, and assigned to procedure-level splits to prevent leakage.

We organize the interactions into five temporal modes: real-time perception (RT), backward memory (BM), forward proactive QA (FP), proactive alert (PA) and caption memory (CM). Across these modes, tasks cover five clinical scenarios: workflow (S1), instrument and action (S2), anatomy (S3), documentation (S4) and narration (S5). Current-scene questions use labels at the query time. Past labels support queries on durations, first appearances, counts and order. Phase and triplet labels yield summaries and operative notes.

Proactive targets pair an expected answer with an evidence time and a deadline. Phase-transition alerts follow annotated transitions, while preparation reminders precede the next phase. This construction supports joint evaluation of answer content and response timing.

RT and BM use accuracy or task-specific tolerances. FP timing accuracy (T-Acc) checks the first reply after the query for correct content within $[a_j, d_j]$. PA timing F1 (T-F1) uses one-to-one temporal matching with the same requirements. CM uses structured F1 (S-F1) over phase intervals and triplet events. Exact vocabulary matching avoids an LLM judge. Avg_{M} and Avg_{S} average mode and scenario scores, each computed as the mean of its task scores.

3. EXPERIMENTS

3.1. Dataset

SurgStream-Bench uses Cholec80 (80 videos, 7 phases and 7 tools) and CholecT50 (50 videos with phase labels and 100 triplet classes), sampled at 1 fps. After resolving shared recordings, the training, validation and test splits contain 53/11/21 procedures spanning 32.5/6.1/15.5 h, respectively. The benchmark contains 11,667 interaction instances (7,315/1,320/3,032 across the splits), including 1,893 FP/PA instances.

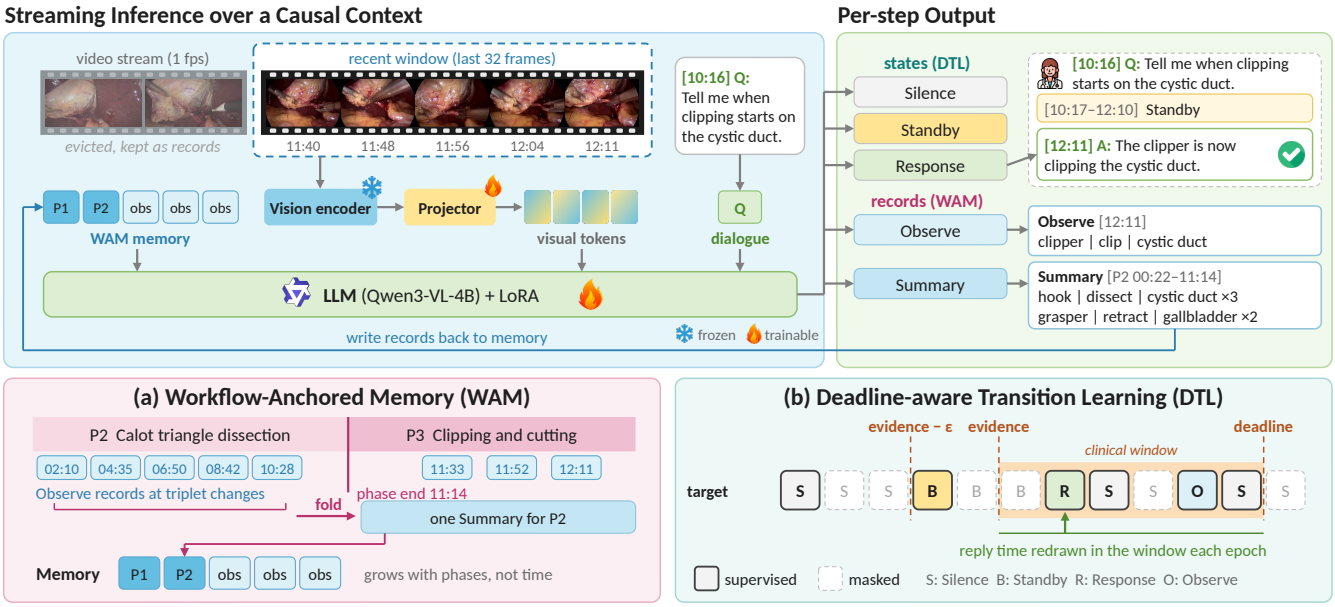


Fig. 2. Overview of SurgStream. At each step, the LLM reads WAM records, the recent visual window and pending dialogue, and emits a control state with optional text. (a) WAM writes *Observe* records at triplet changes and folds them into a *Summary* at each phase boundary. (b) DTL redraws the response time within the clinical window in each epoch and supervises states on a sparse mask.

Table 1. Comparison with state-of-the-art methods on SurgStream-Bench. †: fine-tuned on the SurgStream-Bench training split. The best and second-best results are in **bold** and underlined, respectively. All scores are reported in %.

Methods	Size	RT Acc↑	BM Score↑	FP T-Acc↑	PA T-F1↑	CM S-F1↑	Avg_M ↑	S1	S2	S3	S4	S5	Avg_S ↑
Qwen3-VL [8]	4B	41.73	29.93	0.43	0.30	4.26	15.33	12.11	30.13	46.95	0.68	8.52	19.68
Qwen3.5 [18]	4B	42.70	27.77	2.69	1.08	5.98	16.04	13.06	28.88	43.46	5.38	7.93	19.74
Gemma 4 [19]	E4B	42.89	19.88	0.86	1.60	6.62	14.37	14.17	13.06	52.39	4.64	9.08	18.67
MedGemma [20]	4B	34.54	11.14	0.43	0.73	3.48	10.06	8.84	11.04	36.66	0.89	5.17	12.52
Hulu-Med [10]	4B	42.77	29.70	0.00	0.78	5.95	15.84	13.18	30.15	42.23	6.01	7.28	19.77
MedVLM [21]	7B	55.35	27.80	0.54	1.84	2.43	17.59	13.75	33.36	53.13	0.68	4.85	21.16
EndoChat [22]	13B	39.62	27.92	0.00	0.21	0.86	13.72	5.29	29.57	55.37	2.33	0.48	18.61
Qwen3-VL† [8]	4B	82.04	52.33	21.84	**24.37**	20.68	40.25	35.04	62.50	71.87	22.95	26.73	43.82
OneStreamer† [17]	4B	83.27	59.10	10.87	14.78	15.16	36.64	32.99	62.72	74.81	18.49	24.47	42.70
SurgStream	4B	**84.07**	**64.13**	**23.13**	16.64	**25.22**	**42.64**	**36.84**	**67.70**	**78.06**	**31.45**	**28.79**	**48.57**

Table 2. Ablation study of SurgStream on SurgStream-Bench.

WAM	DTL	RT↑	BM↑	FP↑	PA↑	CM↑	Avg_M ↑	Avg_S ↑
		83.27	59.10	10.87	14.78	15.16	36.64	42.70
✓		82.74	62.46	14.53	17.58	22.84	40.03	46.74
	✓	83.48	61.97	20.23	14.37	22.34	40.48	45.96
✓	✓	84.07	64.13	23.13	16.64	25.22	42.64	48.57

Phase-transition alerts use $[b_k, b_k + 10\,\text{s}]$, and preparation reminders use $[b_k - 60\,\text{s}, b_k]$. Duration, first-appearance and count answers use tolerances of 20%, 10 s and ±1, respectively. Progress questions use S-F1. CM matches phase boundaries and triplet events within 30 s and requires narration within 10 s after each phase.

3.2. Implementation Details

We fine-tune OneStreamer-4B [17], built on Qwen3-VL-4B [8], using LoRA (rank 16) on NVIDIA RTX 6000 Ada GPUs. Stage 1 trains memory writing for 1 epoch, followed by 2 epochs of joint training across all interaction tasks. We use AdamW with an initial learning rate of 10^{-4}, cosine decay and a global batch size of 32. We set $W = 32$, $\kappa = 3\,\text{s}$, $\epsilon = 5\,\text{s}$ and $\eta = 10\,\text{s}$, and use greedy decoding. Each FP/PA instance and *Summary* write point receives two additional training windows. We drop memory records, jitter timestamps and replace elements.

For fair comparison, OneStreamer† uses the same training data, window sampling and epochs. Offline VLMs are adapted to the streaming benchmark through causal frame sampling and periodic querying, with model-specific visual input formatting. All methods are evaluated using the same task definitions, temporal matching criteria, and scoring rules.

3.3. Comparison with State-of-the-Art Methods

We compare SurgStream with zero-shot general offline VLMs [8, 18, 19] and medical VLMs [20, 10, 21, 22], alongside Qwen3-VL† and OneStreamer† [17] fine-tuned on the benchmark training split. As shown in Table 1, SurgStream achieves the best Avg_M of 42.64% and Avg_S of 48.57%, exceeding OneStreamer† by 6.00 and 5.87 percentage points, respectively. The largest mode gains over this matched baseline are on FP (+12.26 points) and CM (+10.06 points). Qwen3-VL† reaches 40.25% Avg_M and leads on PA. Zero-shot VLMs stay

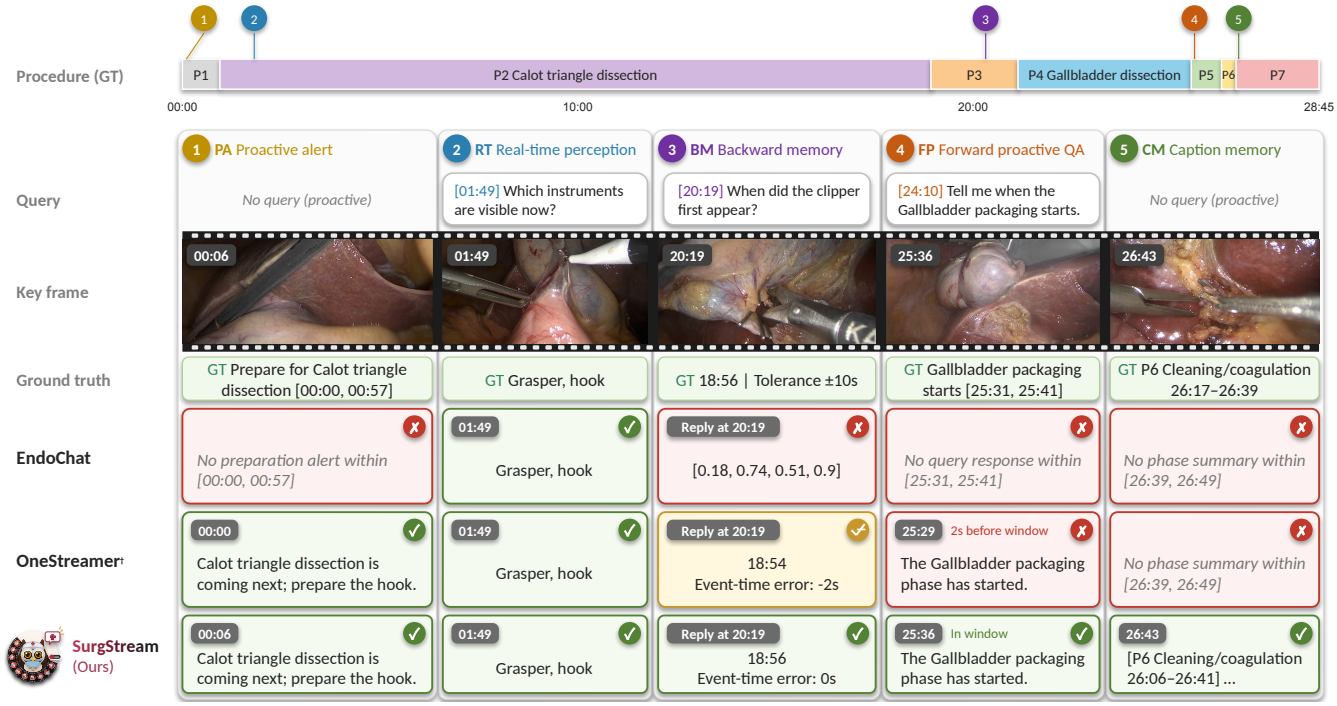


Fig. 3. Qualitative comparison of EndoChat, OneStreamer† and SurgStream on one test procedure across five temporal modes. Responses are excerpted. BM compares event-time estimates (±10 s tolerance), and the CM checkmark denotes a matched phase record.

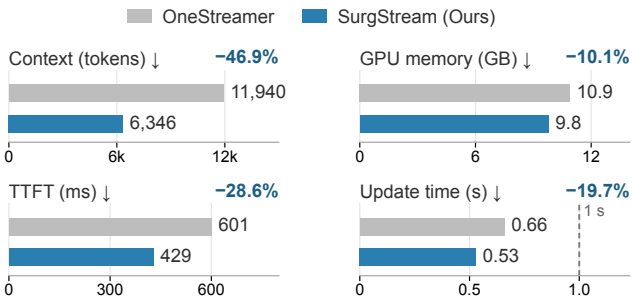


Fig. 4. Efficiency comparison of an unweighted SurgStream checkpoint and zero-shot OneStreamer on two full-length test procedures at 1 fps. Update time is averaged over video steps. GPU memory is measured in GiB.

below 3% on FP and PA, indicating limitations in timely proactive interaction, and medical VLMs do not consistently outperform general ones. Across scenarios, SurgStream improves all five over OneStreamer† and gains most on documentation (S4, +12.96 points). Figure 3 illustrates accurate clipper recall and a valid phase-start response, whereas OneStreamer† replies prematurely. These comparisons support surgical memory and timely interaction.

3.4. Efficiency Analysis

To evaluate streaming efficiency, we compare SurgStream with OneStreamer on two full-length test procedures at 1 fps (Fig. 4). SurgStream reduces peak context, median question-call TTFT and peak GPU memory by 46.9%, 28.6% and 10.1%, respectively. WAM summarizes completed-phase observations, reducing historical text while retaining the recent visual window. Mean update time is 0.53 s excluding frame I/O, below the 1 s input interval. These results demonstrate the inference efficiency of SurgStream when processing full-length surgical videos as continuous streams.

3.5. Ablation Study

We ablate WAM and DTL against OneStreamer† (Table 1), which uses free-caption memory and sparse state supervision (PSTL) [17], with the same data and training schedule. As shown in Table 2, introducing WAM improves BM by 3.36 and CM by 7.68 percentage points, DTL improves FP by 9.36 and CM by 7.18 points, and combining both achieves the best Avg_M of 42.64% and Avg_S of 48.57%. WAM brings the larger gain on BM and PA, whereas DTL brings the larger gain on FP. The combination performs best on all modes except PA, where WAM alone is higher (17.58% *vs.* 16.64%). Together, WAM and DTL provide complementary memory and timing improvements, yielding the best aggregate performance. Still, FP and PA stay below 24% for all variants, so response timing remains the main difficulty.

4. CONCLUSION

We introduced SurgStream for streaming surgical video understanding. Workflow-Anchored Memory (WAM) consolidates timestamped observations into phase summaries, providing compact, queryable surgical memory. Deadline-aware Transition Learning (DTL) resamples target response times within valid clinical windows to guide timely proactive interaction. Experiments on SurgStream-Bench show gains in memory and forward proactive QA with lower context and inference costs. Proactive timing remains challenging, and evaluation is currently limited to laparoscopic cholecystectomy. Future work will improve response timing and evaluate generalization across a broader range of surgical procedures.

5. COMPLIANCE WITH ETHICAL STANDARDS

This research study was conducted retrospectively using human subject data made available in open access by Cholec80 [3] and CholecT50 [23]. Ethical approval was not required as confirmed by the licenses attached with the open access data.

6. ACKNOWLEDGEMENTS

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